



VP AIR TEST NEET (2025-26)

PART TEST - 07

DROPPER

DURATION: 180 Minutes

Date: 21-12-2025

M. MARKS: 720

Topics Covered

Physics:	Electrostatic Potential and Capacitance, Current Electricity
Chemistry:	Hydrocarbons, Haloalkanes and Haloarenes
Botany:	Morphology of Flowering Plants, Anatomy of Flowering Plants
Zoology:	Animal Kingdom

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

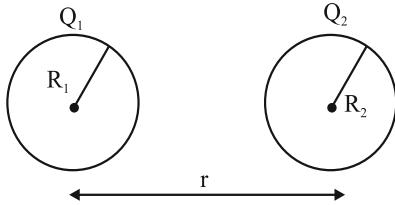
Name of the Student (In CAPITALS): _____

Roll Number: _____

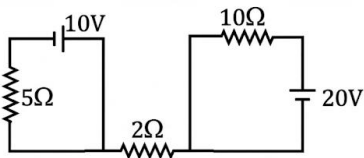
OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. Two non-conducting hollow uniformly charged spheres of radii R_1 and R_2 with charges Q_1 and Q_2 respectively are placed such that the distance between their centers is r (where $r > R_1 + R_2$). Which of the following expressions represents the total electrostatic energy of the system?



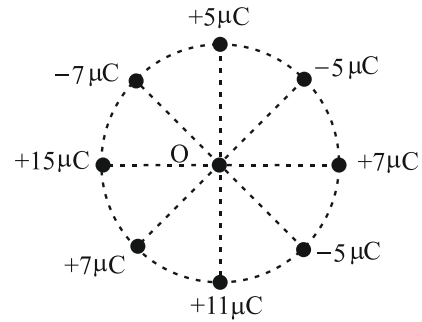
- (1) $\frac{kQ_1Q_2}{r}$
(2) $\frac{kQ_1^2}{2R_1} + \frac{kQ_2^2}{2R_2}$
(3) $\frac{kQ_1^2}{2R_1} + \frac{kQ_2^2}{2R_2} + \frac{kQ_1Q_2}{r}$
(4) $\frac{kQ_1^2}{R_1} + \frac{kQ_2^2}{R_2} + \frac{kQ_1Q_2}{r}$
2. In the figure shown the current through 2Ω resistor is



- (1) 2 A
(2) 0 A
(3) 4 A
(4) 6 A
3. An electric dipole with dipole moment p is placed in a uniform electric field E . An external mechanical agent applies a constant torque $\tau_{ext} = \frac{pE}{2}$ on the dipole in a direction opposite to the torque exerted by the electric field. Consider the angle θ between $\vec{p}$ and $\vec{E}$ in the range $0^\circ < \theta < 180^\circ$. The system has two equilibrium positions at angles:

- (1) 30° and 90°
(2) 45° and 60°
(3) 45° and 150°
(4) 30° and 150°

4. Eight charges having the values as shown are arranged symmetrically on a circle of radius 0.4 m in air. Potential at centre O will be

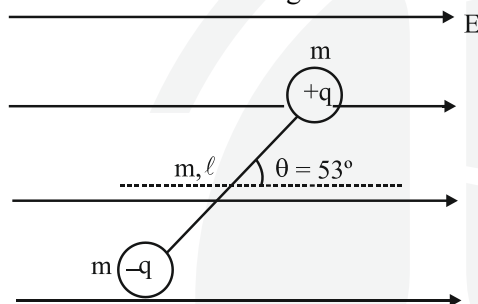


- (1) 63×10^4 volt
(2) 63×10^5 volt
(3) 63×10^6 volt
(4) Zero
5. A galvanometer coil has a resistance 90Ω and full-scale deflection current 10 mA. A 910Ω resistance is connected in series with the galvanometer to make a voltmeter. If the least count of the voltmeter is 0.1 V, the number of divisions on its scale is
- (1) 90
(2) 91
(3) 100
(4) 150
6. At a point $(x, 0, 0)$ in 3D space the electric potential V is $\left(\frac{1000}{x} + \frac{1500}{x^2} + \frac{500}{x^3}\right)$ volt, then electric field at $x = 1$ m will be:
- (1) $-5500(\hat{j} + \hat{k})$ V/m
(2) $-5500\hat{i}$ V/m
(3) $\frac{5500}{\sqrt{2}}(\hat{j} + \hat{k})$ V/m
(4) $\frac{5500}{\sqrt{2}}(\hat{i} + \hat{k})$ V/m
7. Two bulbs rated (25 W, 220 V) and (100 W, 220 V) are connected in series to a 400 V line. Which one is likely to fuse?
- (1) 25 W bulb
(2) 100 W bulb
(3) Both bulbs
(4) No bulb will fuse
8. n small drops of same size have potential V volt each. If they coalesce to form a single large drop, then its potential will be
- (1) $n^{\frac{2}{3}}V$
(2) $n^{\frac{1}{3}}V$
(3) $\frac{V}{n}$
(4) $\frac{V}{\sqrt{n}}$

9. The charge flowing in a conductor varies with time as $Q = t - t^2$ (Where Q is in coulomb and t is in seconds). Then, the current i ;
Identify the **correct** statement (s) from the options given below:

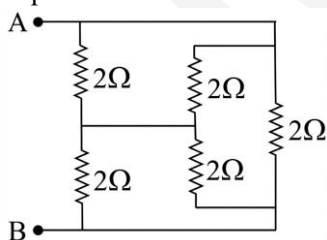
- A. has an initial value equal to 1 Amp.
B. falls to zero after time $t = \frac{1}{2}$ s.
C. decreases linearly with time in interval $t = 0$ to $t = \frac{1}{2}$ s.
(1) A, B and C (2) A and C only
(3) A and B only (4) B and C only

10. Two point masses, each of mass m , with equal and opposite charges of magnitude q are attached to the corners of a non-conducting uniform rod also of mass m and length ℓ . The system is released from rest in a uniform electric field E from an orientation $\theta = 53^\circ$ as shown in the figure.



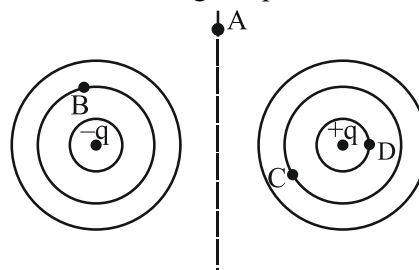
What will be the angular velocity of the rod when it passes through the stable equilibrium position.

- (1) $\sqrt{\frac{24qE}{35ml}}$ (2) $\sqrt{\frac{48qE}{35ml}}$
(3) $\sqrt{\frac{12qE}{5ml}}$ (4) $\sqrt{\frac{6qE}{7ml}}$
11. Find the equivalent resistance across AB

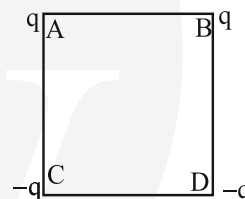


- (1) 1Ω (2) 2Ω
(3) 3Ω (4) 4Ω
12. When a current of 4 A flows within a battery from its positive to negative terminal, the potential difference across the battery is 12 volts. The potential difference across the battery is 9 volts when a current of 2 A flows within it from its negative to its positive terminal. The internal resistance and the e.m.f. of the battery are:
- (1) $0.1\Omega, 4\text{ V}$ (2) $0.2\Omega, 5\text{ V}$
(3) $0.5\Omega, 10\text{ V}$ (4) $0.7\Omega, 10\text{ V}$

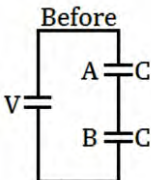
13. Figure shows equi-potential surfaces for a two charges system. At which of the labelled points will a proton have the highest potential energy?



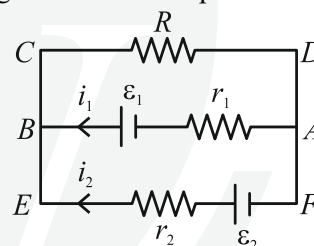
- (1) Point A (2) Point B
(3) Point C (4) Point D
14. If the length of the filament of a heater connected to a constant voltage supply is reduced by 10%, the power of the heater will: (other parameters remain constant)
- (1) increase by about 9%
(2) increase by about 11%
(3) increase by about 19%
(4) decrease by about 10%
15. Four charges are placed on the vertices of a square as shown. Let E be the electric field and V the potential at the centre. If the charges on A and B are interchanged with those on D and C respectively, then –



- (1) $\vec{E}$ remains unchanged, V changes
(2) Both $\vec{E}$ and V changes
(3) $\vec{E}$ and V remains unchanged
(4) $\vec{E}$ changes, V remains unchanged
16. Five parallel plate capacitor have plate areas and plate separations as given below:
Capacitor 1 : Area A_0 , separation d_0
Capacitor 2 : Area $2A_0$, separation $2d_0$
Capacitor 3 : Area $2A_0$, separation $\frac{d_0}{2}$
Capacitor 4 : Area $\frac{A_0}{2}$, separation $2d_0$
Capacitor 5 : Area A_0 , separation $\frac{d_0}{2}$
Rank these according to their capacitances, least to greatest:
- (1) $1 < 2 < 3 < 4 < 5$
(2) $5 < 4 < 3 = 2 < 1$
(3) $4 < 2 = 1 < 3 < 5$
(4) $4 < 1 = 2 < 5 < 3$

17. A capacitor is connected to a battery. The force of attraction between the plates when the separation between them is halved while connected to the battery
- remains the same
 - becomes eight times
 - becomes four times
 - becomes two times
18. A current I flows through a uniform wire of diameter d when the electron drift velocity is v . The same current will flow through a wire of diameter $\frac{d}{2}$ made of the same material then the drift velocity of the electrons is;
- $\frac{v}{4}$
 - $\frac{v}{2}$
 - $2v$
 - $4v$
19. A capacitor of capacitance C is initially charged to a potential difference of V volt. Now it is connected to a battery of $2V$ volt with opposite polarity. The ratio of heat generated to the final energy stored in the capacitor will be
- 1.75
 - 2.25
 - 2.5
 - $\frac{1}{2}$
20. Two identical capacitors are connected in series as shown in the figure. A dielectric slab ($K > 1$) is placed between the plates of the capacitor B and the battery remains connected. Select **correct** statement.
- 
- The charge supplied by the battery increases.
 - The capacitance of the system decreases.
 - The electric field in the capacitor B increases.
 - The electrostatic potential energy decreases.
21. An uncharged capacitor of capacitance $4 \mu F$, a battery of emf 12 volt and a resistor of $2.5 M\Omega$ are connected in series. The time after which $V_c = 3V_R$ is (Take $\ln 2 = 0.693$)
- 6.93 sec.
 - 13.86 sec.
 - 20.52 sec.
 - 10.53 sec.
22. 10^{20} electrons are flowing through a cross section of a wire every second. The current in the wire is;
- 1.6×10^{-19} A
 - 1 A
 - 16 A
 - 10 A

23. **Statement I:** When no current flows through a conductor, the average speed of a free electron over a large period of time is zero.
Statement II: The average velocity of a free electron over a large period of time is zero, when no current flows through a conductor.
- Statement I is correct but Statement II is incorrect.
 - Statement I is incorrect but Statement II is correct.
 - Both Statement I and Statement II are correct.
 - Both Statement I and Statement II are incorrect.
24. A piece of Copper and another of Germanium are cooled from room temperature to 80 K. The resistance of:
- Both of them increases.
 - Both of them decreases.
 - Copper increases and that of Germanium decreases.
 - Copper decreases and that of Germanium increases.
25. In the electrical circuit shown in the figure, which of the following option is the **correct** equation on applying Kirchhoff's loop law?



- $\epsilon_1 - (i_1 + i_2)R + i_1 r_1 = 0$
 - $\epsilon_1 - (i_1 + i_2)R - i_1 r_1 = 0$
 - $\epsilon_2 - i_2 r_2 + \epsilon_1 - i_1 r_1 = 0$
 - $-\epsilon_2 - (i_1 + i_2)R + i_2 r_2 = 0$
26. The electric potential at a point due to an electric dipole with dipole moment $\vec{p}$ at a position vector $\vec{r}$ from the center of dipole is;
- $K \frac{\vec{p} \cdot \vec{r}}{r^3}$
 - $K \frac{\vec{p} \cdot \vec{r}}{r^2}$
 - $2K \frac{\vec{p} \cdot \vec{r}}{r^3}$
 - $2K \frac{\vec{p} \cdot \vec{r}}{r^2}$
27. The separation between the plates of an isolated charged parallel plate capacitor is increased. Which of the following quantities will **not** change?
- charge on the capacitor.
 - potential difference across the capacitor.
 - energy stored in the capacitor.
 - Both (2) and (3)

28. A heater coil is rated 100 W, 220 V. It is cut into two identical parts. Both parts are connected together in parallel to the common source of 220 V. The energy liberated per second in the new combination is:

(1) 50 W (2) 100 W
(3) 200 W (4) 400 W

29. For a uniform non-conducting solid sphere of radius R and charge q , match the description in **List-I** with the correct expression in **List-II**.

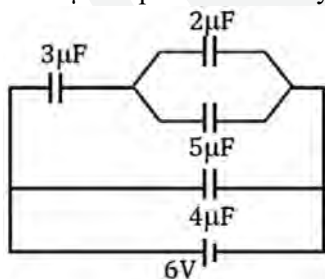
List-I		List-II	
A.	Electric potential at a distance ' r ' from the center such that $r > R$	I.	$\frac{q}{4\pi\epsilon_0 R}$
B.	Electric potential on the surface of the sphere	II.	$\frac{q}{4\pi\epsilon_0 2R^3} [3R^2 - r^2]$
C.	Electric potential at a distance ' r ' from the center such that $r < R$	III.	$\frac{q}{4\pi\epsilon_0 r}$

(1) A-I, B-II, C-III (2) A-II, B-I, C-III
(3) A-III, B-I, C-II (4) A-III, B-II, C-I

30. Two points are at a distance a and b ($a < b$) from a long wire of charge per unit length λ . The potential difference between the points is proportional to;

(1) $\frac{b}{a}$ (2) $\frac{b^2}{a^2}$
(3) $\sqrt{\frac{b}{a}}$ (4) $\log_e \left(\frac{b}{a} \right)$

31. In the circuit shown in figure, the ratio of charges on $5 \mu F$ and $4 \mu F$ capacitor at steady state is:



(1) 4:5 (2) 3:5
(3) 3:8 (4) 1:2

32. In which of the following orientation of electric dipole placed in uniform electric field, it is in unstable equilibrium?

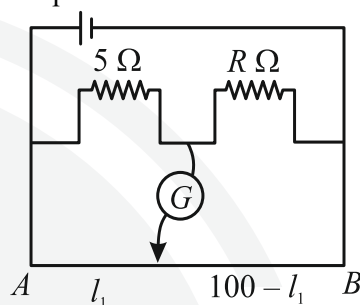
(1)

(2)

(3)

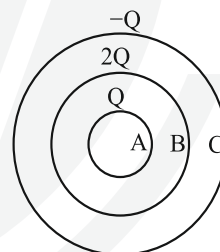
(4)

33. In the circuit given below the resistances in the two arms of the meter bridge are 5Ω and R , respectively when the balance point is at l_1 from A. When the resistance R is shunted with an equal resistance in parallel, the new balance point is at $1.6l_1$ from point A. The value of resistance R is:



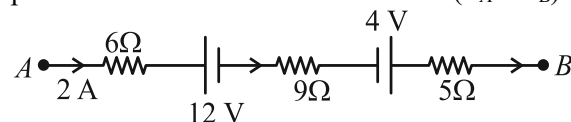
(1) 10Ω (2) 15Ω
(3) 20Ω (4) 25Ω

34. Charge Q , $2Q$ and $-Q$ are given to three concentric conducting spherical shells A, B and C respectively. The ratio of charges on the inner and the outer surfaces of the shell 'C' will be :-



(1) $\frac{3}{4}$ (2) $-\frac{3}{4}$
(3) $\frac{3}{2}$ (4) $-\frac{3}{2}$

35. For the part of a circuit shown in the figure, the potential difference between A and B ($V_A - V_B$) is:



(1) 24 V (2) 14 V
(3) 32 V (4) 48 V

36. The work done in carrying a proton from one point to another point is $3.2 \times 10^{-10} \text{ J}$. The potential difference between the two points is;

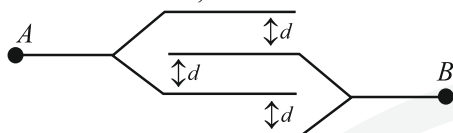
(1) $2 \times 10^9 \text{ V}$ (2) 10^9 V
(3) 10^{12} V (4) 10^{15} V

37. Which of the following statements is **correct** regarding electrostatics of conductors?

- A. The interior of a conductor with no cavity can have no excess charge in the static situation.
- B. Electrostatic potential is constant throughout the volume of the conductor.
- C. Electrostatic potential has same value inside as that on its surface.

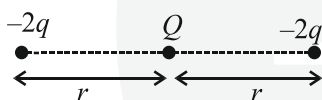
- (1) A, B, C only (2) A, B only
(3) B, C only (4) A, C only

38. If area of each plate is A and distance between the plates is d then net capacitance of the arrangement between A and B is;



- (1) $\frac{\epsilon_0 A}{3d}$ (2) $\frac{2\epsilon_0 A}{d}$
(3) $\frac{3\epsilon_0 A}{d}$ (4) $\frac{4\epsilon_0 A}{d}$

39. Three charges $-2q$, Q and $-2q$ are placed on a straight line as shown in figure. If the total potential energy of the system is zero then the value of $\frac{Q}{q}$ will be;



- (1) $\frac{1}{3}$ (2) $\frac{3}{5}$
(3) $\frac{1}{2}$ (4) $\frac{2}{5}$

40. The capacitance of a charged isolated spherical capacitor is 10 pF. The radius of capacitor is:

- (1) 3 cm (2) 6 cm
(3) 4 cm (4) 9 cm

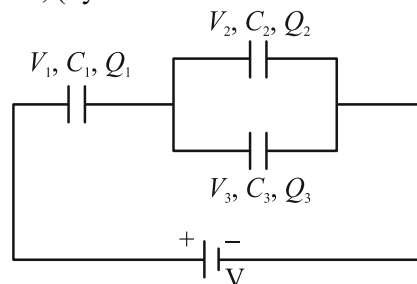
41. Three point charges q , q and $-2q$ are placed at the corners of an equilateral triangle of side L . Calculate the work done by external force in moving all the charges slowly to infinity is:

- (1) $\frac{1}{4\pi\epsilon_0} \frac{3q^2}{L}$ (2) $-\frac{1}{4\pi\epsilon_0} \frac{3q^2}{L}$
(3) $\frac{1}{4\pi\epsilon_0} \frac{5q^2}{L}$ (4) $-\frac{1}{4\pi\epsilon_0} \frac{5q^2}{L}$

42. If a wire of resistance R is compressed such that its new length becomes 80% of its original length, then resistance of wire will decrease by:

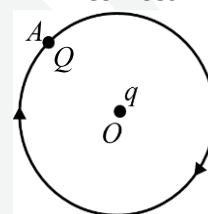
- (1) 64% (2) 36%
(3) 20% (4) 80%

43. In an adjoining figure shown, three capacitors C_1 , C_2 and C_3 joined to a battery. The **correct** relation will be; (Symbols have their usual meanings).



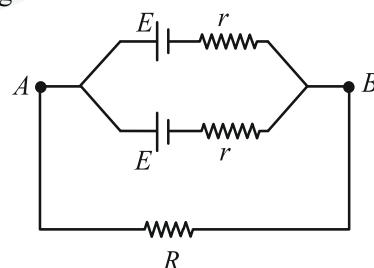
- (1) $Q_1 = Q_2 = Q_3$ and $V_1 = V_2 = V_3 = V$
(2) $Q_1 = Q_2 + Q_3$ and $V = V_1 + V_2 + V_3$
(3) $Q_1 = Q_2 + Q_3$ and $V = V_1 + V_2$
(4) $Q_2 = Q_3$ and $V = V_2 + V_3$

44. A charge q is placed at the centre (O) of a circle. Another charge Q is to be slowly taken from point A along the circle and finally back to point A as shown. Choose the **correct** statement:



- (1) Work done by external force in moving Q on complete circle is positive.
(2) Work done by external force in moving Q on complete circle is negative
(3) Work done by external force in moving Q on complete circle is zero.
(4) Data insufficient

45. If two cells of equal emf E each and internal resistance r each are combined as shown, then what should be the value of emf (E_{eq}) and internal resistance (r_{eq}) of a single cell which can replace the above two cells without affecting current through R ?

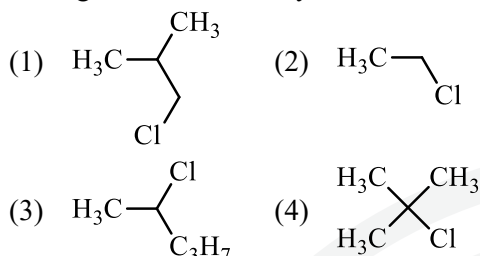


- (1) $E_{eq} = \frac{E}{2}$, $r_{eq} = \frac{r}{2}$
(2) $E_{eq} = E$, $r_{eq} = \frac{r}{2}$
(3) $E_{eq} = 2E$, $r_{eq} = r$
(4) $E_{eq} = \frac{E}{2}$, $r_{eq} = r$

46. Acid catalyzed hydration of alkene is an example of:

- (1) nucleophilic addition reaction.
- (2) electrophilic addition reaction.
- (3) electrophilic substitution reaction.
- (4) nucleophilic substitution reaction.

47. Among the following, compound that most readily undergoes substitution by S_N2 mechanism is:



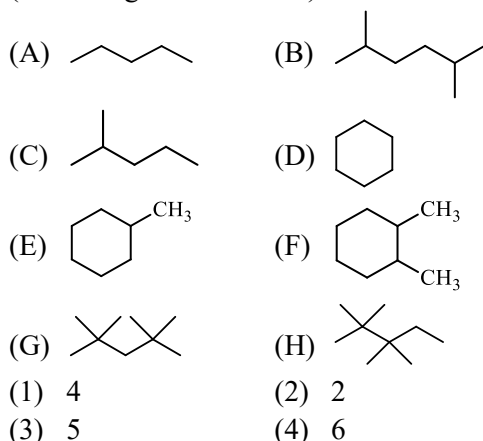
48. An organic compound (A) $C_7H_{15}Cl$ on treatment with alcoholic caustic potash gives a hydrocarbon (B) C_7H_{14} . (B) on treatment with ozone and subsequent hydrolysis gives acetone and butyraldehyde. What are (A) and (B) respectively?

- (1) A: $(CH_3)_2CClCH_2CH_2CH_2CH_3$ and B: $(CH_3)_2C=CHCH_2CH_2CH_3$
- (2) A: $(CH_3)_2CHCHClCH_2CH_2CH_2$ and B: $(CH_3)_2CHCH_2CH_2CH=CH_2$
- (3) A: $(CH_3)_2CClCH_2CH_2CH_2CH_3$ and B: $(CH_3)_2CHCH_2CH_2CH=CH_2$
- (4) A: $(CH_3)_2CHCHClCH_2CH_2CH_3$ and B: $CH_3CH=CHCH_2CH_2CH_3$

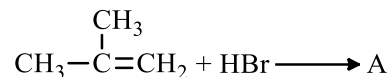
49. The **correct** order of rate of reaction of alkanes with halogens is:

- (1) $F_2 > Cl_2 > Br_2 > I_2$
- (2) $Cl_2 > Br_2 > I_2 > F_2$
- (3) $I_2 > Br_2 > Cl_2 > F_2$
- (4) $F_2 > I_2 > Br_2 > Cl_2$

50. Number of alkanes that gives four alkyl chloride on monochlorination among the following are: (Excluding stereoisomers)

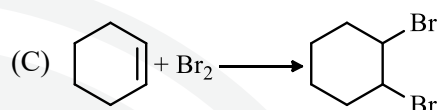


51. The major product A obtained in the given reaction is:



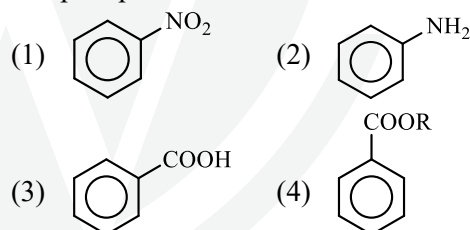
- (1) 2-Bromo-2-methylpropane
- (2) 1-Bromo-2-methylpropane
- (3) 2-Bromobutane
- (4) 1-Bromobutane

52. Which of the following statements is **correct** for the given reactions:



- (1) (A) and (B) are elimination reactions and (C) is an addition reaction.
- (2) (A) is elimination, (B) is substitution and (C) is addition reaction.
- (3) (A) is elimination, (B) and (C) are substitution reactions.
- (4) (A) is substitution, (B) and (C) are addition reactions.

53. In which of the following compounds the electrophile will be directed to be attached on ortho and para positions?



54. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

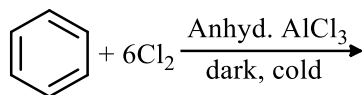
Assertion A: The staggered conformation of ethane is more stable than the eclipsed conformation of ethane.

Reason R: The staggered form has the least torsional strain and the eclipsed form has the maximum torsional strain.

In the light of the above statements, choose the **correct** answer from the options given below:

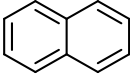
- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

55. Benzene on treatment with excess of chlorine in the presence of anhydrous AlCl_3 forms:

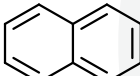


- (1) Hexachlorobenzene
- (2) Chlorobenzene
- (3) Benzene hexachloride
- (4) Cyclohexyl chloride

56. Which of the following compound is non-aromatic?

- (1) 
- (2) 
- (3) 
- (4) 

57. Match the **List-I** with **List-II**.

List-I		List-II	
(A)		(I)	Aromatic and has 10π electrons
(B)		(II)	Anti-aromatic and has 4π electrons
(C)		(III)	Aromatic and has 6π electrons
(D)		(IV)	Non-aromatic and has 4π electrons

Choose the **correct** answer from the options given below;

- (1) A-III, B-I, C-IV, D-II
 - (2) A-III, B-I, C-II, D-IV
 - (3) A-I, B-III, C-II, D-IV
 - (4) A-III, B-II, C-IV, D-I
58. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Aryl halides are highly reactive towards nucleophilic substitution reaction.
Reason R: In aryl halides, C—Cl bond acquires a partial double bond character.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false
 - (2) A is false but R is true
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is NOT the correct explanation of A.

59. Consider the statements given below for the conversion of 2-Bromopentane to form Pent-2-ene.

- A. It is β -elimination reaction.
- B. Pent-1-ene is the major product.
- C. It is dehydrohalogenation reaction.
- D. Pent-2-ene is the minor product.

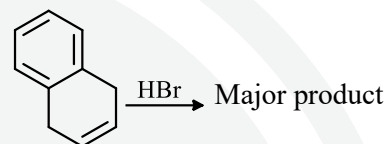
The **correct** statement(s) is/are:

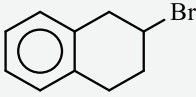
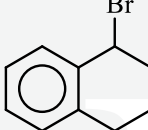
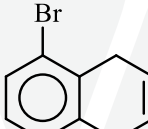
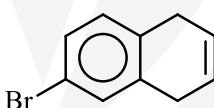
- (1) A, C and D
- (2) B, C and D
- (3) A, B and D
- (4) Only A and C

60. Which of the following is **correct** for acetylene?

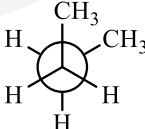
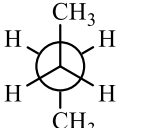
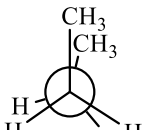
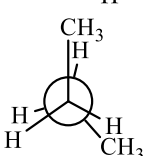
- (1) It is less acidic than neo-pentane.
- (2) It is a solid at room temperature.
- (3) s-character of C—H bond is high in acetylene because it is sp hybridised.
- (4) It is less acidic than But-1-ene.

61. The major product formed in the above reaction is:



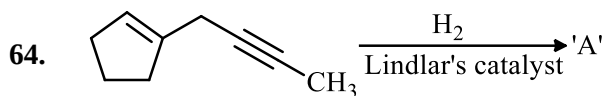
- (1) 
- (2) 
- (3) 
- (4) 

62. In the following the most stable conformation of n-butane is:

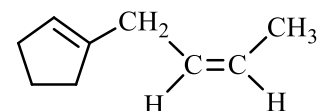
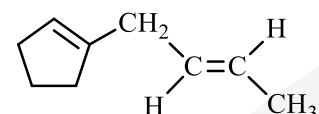
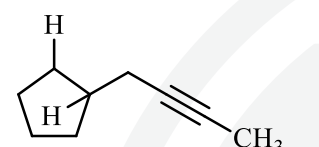
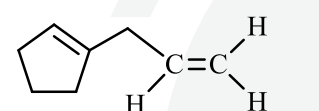
- (1) 
- (2) 
- (3) 
- (4) 

63. Which of the following hydrocarbons is **not** a gas at room temperature?

- (1) Methane
- (2) Butane
- (3) Propane
- (4) Pentane



Product 'A' will be:

- (1) 
- (2) 
- (3) 
- (4) 

65. 2,3-Dimethyl-2-butene can be prepared by heating which of the following compounds with a strong acid?

- (1) $(\text{CH}_3)_2\text{CH}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}=\text{CH}_2$
- (2) $(\text{CH}_3)_3\text{C}-\text{CH}=\text{CH}_2$
- (3) $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_2-\text{CH}_3$
- (4) $(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{CH}=\text{CH}_2$

66. Among the following alkenes

- (I) 1-Butene
- (II) cis-2-Butene
- (III) trans-2-Butene

The decreasing order of stability is:

- (1) II > I > III
- (2) III > II > I
- (3) III > I > II
- (4) I > II > III

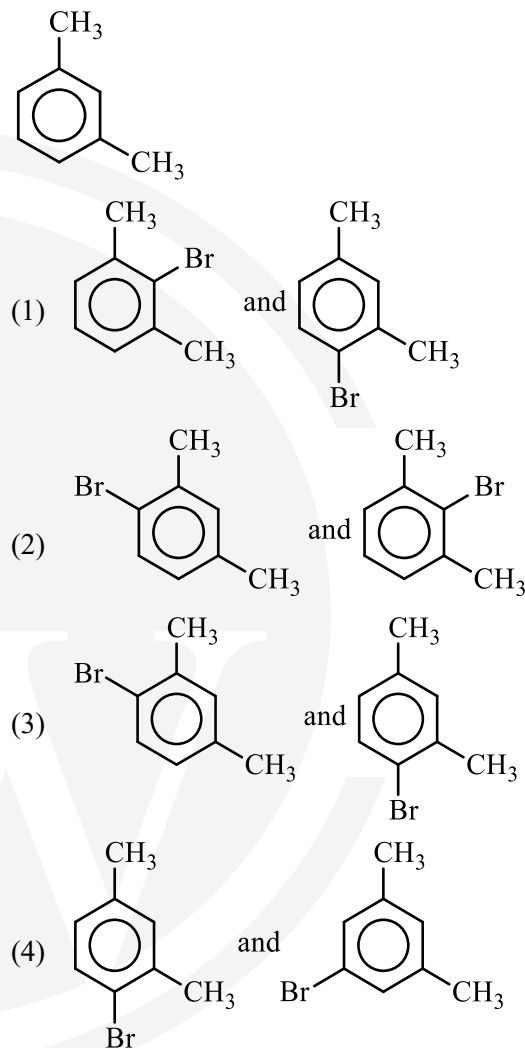
67. Chloroform is slowly oxidised by air in the presence of light to an extremely poisonous gas, known as:

- (1) Freon
- (2) Carbonyl chloride
- (3) DDT
- (4) Dichloromethane

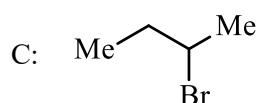
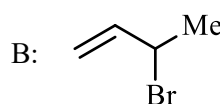
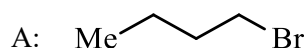
68. Which of the following reagents will be able to distinguish between 1-butyne and 2-butyne?

- (1) NaNH_2
- (2) HCl
- (3) O_2
- (4) Br_2

69. What products are formed when the following compound is treated with Br_2 in the presence of FeBr_3 ?



70. Consider the following bromides:



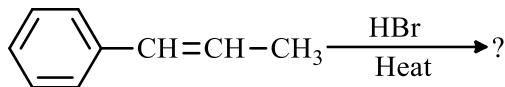
The **correct** order of $\text{S}_{\text{N}}1$ reaction is:

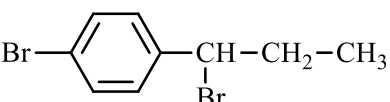
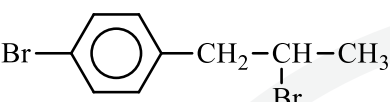
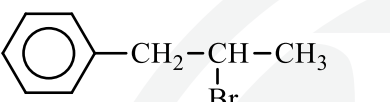
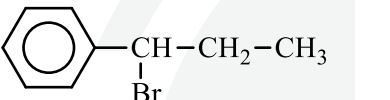
- (1) C > B > A
- (2) A > B > C
- (3) B > C > A
- (4) B > A > C

71. Which one of the following is likely to give a precipitate with AgNO_3 solution?

- (1) $\text{CH}_2=\text{CH}-\text{Cl}$
- (2) CCl_4
- (3) CHCl_3
- (4) $(\text{CH}_3)_3\text{CCl}$

72. The major product in the following conversion is:

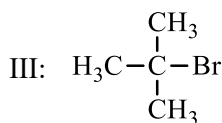
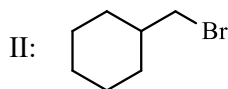
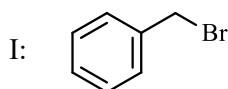


- (1) 
- (2) 
- (3) 
- (4) 

73. When CH_3Cl and anhydrous AlCl_3 are used in Friedel-Crafts reaction, the electrophile generated is:

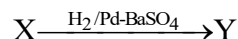
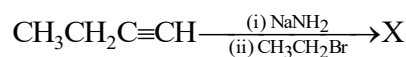
- (1) Cl^+
- (2) AlCl_4^-
- (3) CH_3^+
- (4) CH_3^-

74. For the following compounds, the **correct** statement with respect to nucleophilic substitution reaction is(are):



- (1) Compound II undergoes inversion of configuration
- (2) I follow $\text{S}_{\text{N}}2$ mechanism
- (3) II follow $\text{S}_{\text{N}}1$ mechanism
- (4) III follow $\text{S}_{\text{N}}2$ mechanism

75. Identify Y in the following synthetic scheme and write their structures

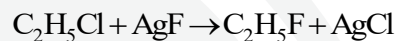


- (1) Hex-2-ene
- (2) But-2-ene
- (3) But-1-ene
- (4) Hex-3-ene

76. A compound X undergoes ozonolysis followed by $\text{Zn}/\text{H}_2\text{O}$ to give only ethane-1,2-dial as the product. The compound X is:

- (1) Cyclobutene
- (2) Cyclobuta-1,3-diene
- (3) But-2-ene
- (4) But-1-ene

77. The above reaction is called:



- (1) Friedel Craft's reaction
- (2) Swart's reaction
- (3) Kolbe's reaction
- (4) Wurtz reaction

78. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

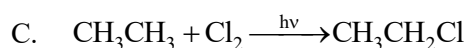
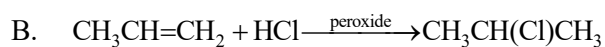
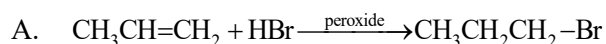
Assertion A: Rate of reaction is dependent only on the concentration of nucleophile in $\text{S}_{\text{N}}1$ reactions.

Reason R: Polar protic solvent favours $\text{S}_{\text{N}}1$ reactions.

In the light of the above statements, choose the **correct** answer from the options given below:

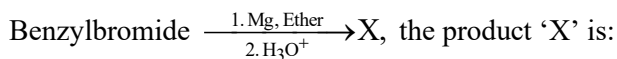
- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

79. Which of the following is/are free radical reaction(s)?



- (1) Only B
- (2) Only A
- (3) A, B and C
- (4) A and C

80. In the following reaction:



- (1) $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$
- (2) $\text{C}_6\text{H}_5\text{CH}_3$
- (3) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{C}_6\text{H}_5$
- (4) $\text{C}_6\text{H}_5\text{CH}_2\text{OCH}_2\text{C}_6\text{H}_5$

81. The number of the sigma (σ) and pi (π) bonds in pent-3-en-1-yne is:

- (1) 10 σ bonds and 3 π bonds.
- (2) 8 σ bonds and 5 π bonds.
- (3) 11 σ bonds and 2 π bonds.
- (4) 13 σ bonds and 3 π bonds.

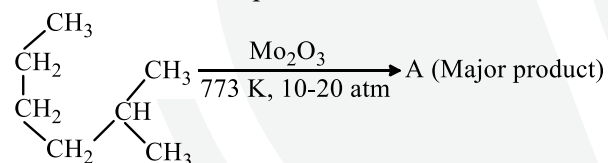
82. When $\text{CH}_3\text{--C}\equiv\text{CH}$ is treated with aqueous H_2SO_4 in presence of HgSO_4 , the major product formed is:

- (1) Butanone
- (2) Propan-2-ol
- (3) Propan-1-ol
- (4) Propanone

83. Which of the following molecules has the highest dipole moment?

- (1) CH_3Cl
- (2) CH_2Cl_2
- (3) CHCl_3
- (4) CCl_4

84. Identify 'A' in the given chemical reaction and choose the **correct** option.

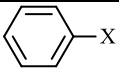
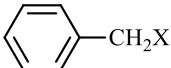


- (1) Benzene
- (2) Cyclohexane
- (3) Methyl cyclohexane
- (4) Toluene

85. 2° alkyl halide among the following is:

- (1) isopropyl chloride.
- (2) isobutyl chloride.
- (3) n-propyl chloride.
- (4) n-butyl chloride.

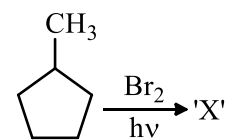
86. Match the **List-I** with **List-II**.

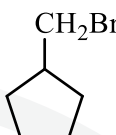
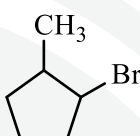
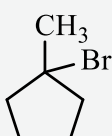
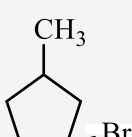
List-I		List-II	
(A)	$\text{CH}_2=\text{CH--CH}_2\text{--X}$	(I)	Benzylic halide
(B)	$\text{CH}_2=\text{CH--X}$	(II)	Allylic halide
(C)		(III)	Vinylic halide
(D)		(IV)	Aryl halide

Choose the **correct** answer from the options given below;

- (1) A-II, B-III, C-IV, D-I
- (2) A-I, B-II, C-IV, D-III
- (3) A-III, B-IV, C-II, D-I
- (4) A-IV, B-I, C-II, D-III

87. In the following reaction, the major product 'X' is:



- (1) 
- (2) 
- (3) 
- (4) 

88. On treating a mixture of two alkyl halides with sodium metal in dry ether, 2-methyl propane was obtained. The alkyl halides are:

- (1) 1-Chloropropane and chloroethane
- (2) 2-Chloropropane and chloroethane
- (3) 2-Chloropropane and chloromethane
- (4) 1-Chloropropane and chloromethane

89. On monochlorination of 2-methyl butane, the total number of chiral compounds is:

- (1) 2
- (2) 4
- (3) 6
- (4) 8

90. Which of the following will have a meso-isomer also?

- (1) 2-Chlorobutane
- (2) 2,3-Dichlorobutane
- (3) 2,3-Dichloropentane
- (4) 2-Hydroxy propanoic acid

91. Which of the following statements are correct regarding a young dicotyledonous root?
- The outermost layer is epiblema, with some cells forming unicellular root hairs.
 - The cortex consists of several layers of thin-walled parenchyma cells with intercellular spaces.
 - The endodermis is comprised of a single layer of barrel-shaped cells without intercellular spaces.
 - The pith is large and well-developed.
 - It usually has more than six xylem bundles.

Choose the *most appropriate* answer from the options given below.

- A, B, C, D and E
- A, B and C only
- D and E only
- A, C and E only

92. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In *Rhizophora*, many roots come out of the ground and grow vertically upwards.

Reason (R): These roots of *Rhizophora*, called stilt roots, help to get oxygen for respiration.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- Both (A) and (R) are true and (R) is the correct explanation of (A).
- Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (A) is true but (R) is false.
- (A) is false but (R) is true.

93. Identify the **correct** statement about the simple permanent tissue in which the cells are usually dead at maturity.

- The cells have thick cell walls that lack pits.
- On the basis of variation in form, structure, origin and development, its cells can be of two types.
- The cell walls are deposited with pectin and cellulose.
- These are found in young stem and petiole.

94. Match **List-I** with **List-II** w.r.t papilionaceous aestivation.

List-I (Petal)		List-II (Description)	
(A)	Standard	(I)	Two and free
(B)	Keel	(II)	One and largest
(C)	Wing	(III)	Two and fused

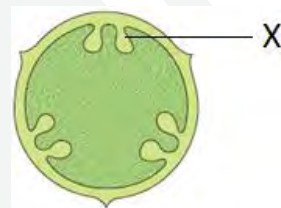
Choose the *most appropriate* answer from the options given below.

- A-II, B-III, C-I
- A-II, B-I, C-III
- A-I, B-II, C-III
- A-III, B-I, C-II

95. The meristems which occur at the tips of roots and shoots;

- disappear during secondary growth.
- are responsible for regenerating the parts in grasses that are removed by the grazing herbivores.
- include the cylindrical meristems.
- produce primary tissues.

96. Shown below is a type of placentation. Choose the correct statement about it.



- The ovary here is one-chambered but becomes two-chambered due to the formation of false placentas.
- The label 'X' refers to the structure that becomes seed after fertilisation.
- Such type of placentation is seen in tomato and lemon.
- In this type of placentation, the placenta develops at the base of ovary.

97. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The outside of the epidermis in plants is often covered with a waxy thick layer called the cuticle.

Reason (R): The cuticle prevents the loss of water, hence present in all epidermal parts of a plant body. In the light of the above statements, choose the *most appropriate* answer from the options given below.

- Both (A) and (R) are true and (R) is the correct explanation of (A).
- Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (A) is true but (R) is false.
- (A) is false but (R) is true.

98. Read the following statements.

- A. A flower having either only stamens or only carpels is bisexual.
- B. When a flower can be divided into two similar halves only in one particular vertical plane, it is zygomorphic.
- C. An asymmetric flower is present in *Datura* and *canna*.
- D. Flowers having reduced leaf found at the base of the pedicel are called stipulate.
- E. When calyx and corolla are not distinct, they are termed as perianth.

In the light of above statements, choose the **correct** answer from the options given below:

- (1) A, B and C are correct
- (2) B, C and E are incorrect
- (3) A, C and D are incorrect
- (4) A, D and E are correct

99. Arrange the following structures of a primary dicotyledonous root from the outermost to the innermost region.

- A. Pith
- B. Cortex
- C. Epidermis
- D. Pericycle
- E. Vascular bundle

Choose the **correct** option.

- (1) C → B → A → D → E
- (2) C → B → D → E → A
- (3) E → D → A → B → C
- (4) B → C → A → D → E

100. Go through the floral formula given below:



Choose the **incorrect** statement about it.

- (1) This is the floral formula for the cotton family.
- (2) Epik refers to the leafy structures present outside calyx.
- (3) The petals are fused.
- (4) The ovary is said to be superior.

101. The 'reductionist biology' approach mainly focussed on;

- (1) the study of living processes.
- (2) description of life forms and their structure.
- (3) observation of organisms through naked eyes.
- (4) observation of organisms through magnifying lenses.

102. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Xylem	(I)	Only one type of cell is living, other 3 are dead at maturity
(B)	Phloem	(II)	The curly-top virus spreads through it
(C)	Collenchyma	(III)	Occurs in layers below the epidermis in most of the dicotyledonous plants
(D)	Parenchyma	(IV)	Forms the major component within plant organs

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-II, D-IV
- (2) A-I, B-III, C-IV, D-II
- (3) A-III, B-I, C-IV, D-II
- (4) A-I, B-II, C-III, D-IV

103. Identify the **correctly** matched pair.

- (1) Alternate phyllotaxy - *Calotropis*
- (2) Opposite phyllotaxy - China rose
- (3) Palmately compound leaf - *Alstonia*
- (4) Pinnately compound leaf - Neem

104. Given below are two statements:

Statement I: In monocotyledonous stems, vascular bundles are conjoint and closed, and are scattered in the ground tissue.

Statement II: In dicotyledonous stems, vascular bundles are arranged in a ring and are conjoint, open.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

- 105.** All the following are the features of china rose, except;
- (1) stamens are united into one bunch or one bundle.
 - (2) presence of unilocular ovary.
 - (3) the gynoecium occupies the highest position while the other parts are situated below it.
 - (4) one margin of the floral appendage overlaps that of the next one and so on.
- 106.** A student observes the transverse section of a plant material which shows that the central vascular bundles were larger than the peripheral ones and had water-containing cavities. The student should identify the material as a:
- (1) Dicot stem.
 - (2) Monocot stem.
 - (3) Dicot root.
 - (4) Monocot root.
- 107.** Read the following statements about the structure of a dicotyledonous seed.
- A. The outermost covering of a seed is the seed coat which has two layers, the outer testa and the inner tegmen.
 - B. The hilum is a scar on the seed coat through which the developing seeds were attached to the fruit.
 - C. Above the hilum is a small pore called the micropyle.
 - D. The cotyledons are often fleshy and full of reserve food materials.
 - E. In endospermic seeds like castor, the endosperm is not present.
- Choose the option with all **correct** statements.
- (1) A, B, C, D and E
 - (2) A, B and C only
 - (3) A, B, C and D only
 - (4) D and E only

- 108.** Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
- Assertion (A):** In stems, the arrangement of primary xylem is called endarch.
- Reason (R):** In stems, the protoxylem lies towards the periphery and the metaxylem lies towards the centre.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (3) (A) is true but (R) is false.
 - (4) (A) is false but (R) is true.
- 109.** Choose the feature that is **not** true for monocot plants.
- (1) The venation in leaves is generally parallel.
 - (2) The roots are always tap roots.
 - (3) They do not show secondary growth.
 - (4) The seed is generally endospermic.
- 110.** Read the following statements.
- A. In monocot stem, vascular bundles are surrounded by a layer of bundle sheath cells.
 - B. In dorsiventral leaf, vascular bundles are surrounded by thin walled bundle sheath cells.
 - C. In dicot leaf, vascular bundles are surrounded by several layers of bundle sheath cells.
 - D. In monocot stem, the bundle sheath cells are sclerenchymatous.
 - E. In isobilateral leaf, vascular bundles are not surrounded by bundle sheath cells.
- Which of the above statements are **incorrect**?
- (1) A, B and D only
 - (2) D and E only
 - (3) A and B only
 - (4) B, C and E only
- 111.** Given below are two statements:
- Statement I:** A simple leaf never has incisions on the lamina.
- Statement II:** A bud is present in the axil of the petiole in both simple and compound leaves, and in the axil of leaflets of the compound leaf.
- In the light of the above statements, choose the correct answer from the options given below:
- (1) Both Statement I and Statement II are correct.
 - (2) Both Statement I and Statement II are incorrect.
 - (3) Statement I is correct but Statement II is incorrect.
 - (4) Statement I is incorrect but Statement II is correct.

112. Identify the **correct** statement.

- (1) Parenchyma cells provide mechanical support to mature organs.
- (2) Collenchyma cells provide mechanical support to the growing parts of the plant such as young stem and petiole of a leaf.
- (3) Sclerenchyma cells provide mechanical support to primary meristems.
- (4) Leaf and stem may be modified as thorns/spines to provide mechanical support for climbing.

113. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Photosynthetic petiole	(I)	<i>Pistia</i>
(B)	Lateral branch with short internodes and a rosette of leaves	(II)	Sweet potato
(C)	Underground stems modified for food storage	(III)	Australian acacia
(D)	Adventitious roots that store food	(IV)	Ginger

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-IV, D-II
- (2) A-III, B-II, C-I, D-IV
- (3) A-I, B-III, C-II, D-IV
- (4) A-II, B-IV, C-III, D-I

114. Observe the given diagram of a plant part.



Choose the **correct** statement about it.

- (1) The shape of 'D' depicted here is typical for isobilateral leaves.
- (2) B, D and E together constitute the stomatal aperture.
- (3) 'A' is a specialised endodermal cell.
- (4) This structure is a part of the epidermal tissue system.

115. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The position of ovary with respect to other parts of flower in plum and rose is same.

Reason (R): In plum and rose, the ovary is said to be inferior.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

116. The trichomes;

- A. are epidermal hairs on stem.
- B. are usually unicellular.
- C. are always unbranched.
- D. may be secretory in function.
- E. may be soft or stiff.

Choose the *most appropriate* answer from the options given below.

- (1) A, B, C and E only
- (2) A, B and D only
- (3) C and E only
- (4) A, D and E

117. What is the correct sequence of structures in a floral diagram from innermost to the outermost.

- (1) Calyx → Corolla → Androecium → Gynoecium
- (2) Gynoecium → Androecium → Corolla → Calyx
- (3) Corolla → Calyx → Androecium → Gynoecium
- (4) Calyx → Androecium → Corolla → Gynoecium

118. The medullary rays;

- (1) are made of parenchymatous cells.
- (2) constitute the complex permanent tissues.
- (3) are present radially between the vascular bundles in a monocot stem.
- (4) are dead cells.

- 119.** Identify the **correct** statements from the following.
- In *Monstera*, adventitious roots are present.
 - The main functions of the root system are absorption of water and minerals, providing anchorage, and synthesis of plant growth regulators.
 - Roots arising from parts of the plant other than the radicle are called adventitious roots.
 - In the majority of dicotyledonous plants, the direct elongation of the radicle leads to the formation of the primary root.
 - Prop roots are present in banyan tree.

Choose the **correct** option.

- A, B, C and D only
 - C, D and E only
 - A, B and E only
 - A, B, C, D and E
- 120.** The spongy parenchyma cells of ground tissue in a dorsiventral leaf;
- are placed on the adaxial side.
 - lack intercellular spaces.
 - are oval or round in shape.
 - are placed parallel to each other.
- 121.** Select the **incorrectly** matched pair w.r.t the general characteristics of members in the given plant families.
- Leguminosae - Fruit is caryopsis
 - Solanaceae - Flower is pentamerous
 - Cruciferae - Ovary is superior
 - Gramineae - Inflorescence is spike of spikelet
- 122.** Given below are two statements:

Statement I: The layer of cells called starch sheath is present on the inner side of pericycle in a dicot stem.

Statement II: The innermost layer of cortex of the dicot stem is in the form of semi-lunar patches of sclerenchyma.

In the light of the above statements, choose the **correct** answer from the options given below:

- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.
- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.

- 123.** In mango, the fruit;
- develop from monocarpellary inferior ovary.
 - develops parthenocarpically.
 - has an outer thin epicarp, a middle fleshy edible mesocarp and an inner stony hard endocarp.
 - lacks a pericarp.

124. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Casparian strips	(I)	Large, empty, colourless cells in the adaxial epidermis of grasses
(B)	Stomata	(II)	Unicellular elongations of epidermal cells
(C)	Bulliform cells	(III)	Root endodermis walls deposited with water-impermeable, waxy material suberin
(D)	Root hairs	(IV)	Regulate the process of transpiration and gaseous exchange

Choose the *most appropriate* answer from the options given below.

- A-II, B-IV, C-I, D-III
 - A-III, B-II, C-I, D-IV
 - A-II, B-I, C-III, D-IV
 - A-III, B-IV, C-I, D-II
- 125.** Which of the following is not true for sunflower?
- Basal placentation
 - Ray florets have half-inferior ovary
 - Member of family Compositae
 - Seeds are source of oil

126. In a dorsiventral leaf;

- the epidermis covers both the upper and lower surfaces.
- the abaxial epidermis generally bears more stomata than the adaxial epidermis.
- the vascular tissue system is absent.
- the veins vary in thickness.
- the upper epidermis is called abaxial and the lower epidermis is called adaxial.

The **correct** statements are;

- A, B and D only.
- A, B, C and D only.
- A, D and E only.
- B, C and E only.

127. The scientist Katherine Esau;
- (1) received the National Medal of Science from President George Bush in 2000.
 - (2) was the first woman to be elected to the National Academy of Sciences.
 - (3) authored the book "*Anatomy of Seed Plants*".
 - (4) authored the book "*Plant Anatomy*" which was called the Webster's of plant biology.

128. A plant having polyadelphous stamens and axillary buds modified into thorns, is;
- (1) *Citrus*.
 - (2) *Euphorbia*.
 - (3) *Salvia*.
 - (4) *Opuntia*.

129. Read the following statements w.r.t monocotyledonous root.
- A. The pith is large and well-developed.
 - B. It has epidermis, cortex, endodermis, and pericycle.
 - C. It usually has polyarch xylem bundles.
 - D. The arrangement of vascular bundles is said to be radial.
 - E. The vascular bundles are said to be open.
- The **correct** statements are;
- (1) A, B, C and D only
 - (2) A, C, D and E only
 - (3) B, C and E only
 - (4) A, B, C, D and E

130. Given below are two statements:
- Statement I:** In racemose type of inflorescence, the main axis continues to grow and the flowers are borne laterally in an acropetal succession.
- Statement II:** In cymose type of inflorescence, the main axis terminates in a flower, hence is limited in growth and flowers are borne in a basipetal order.
- In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both Statement I and Statement II are correct.
 - (2) Both Statement I and Statement II are incorrect.
 - (3) Statement I is correct but Statement II is incorrect.
 - (4) Statement I is incorrect but Statement II is correct.

131. The cells of bast fibres;
- (1) have thin cell walls.
 - (2) are made of collenchyma and parenchyma.
 - (3) are much elongated, unbranched and have pointed, needle like apices.
 - (4) at maturity possess protoplasm and are dead.

132. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Apocarpous condition	(I)	Lotus
(B)	Epiphyllous condition	(II)	Brinjal
(C)	Epipetalous condition	(III)	Lily
(D)	Gamopetalous condition	(IV)	Potato

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-II, D-IV
 - (2) A-I, B-II, C-III, D-IV
 - (3) A-III, B-I, C-IV, D-II
 - (4) A-I, B-III, C-II, D-IV
133. Phloem parenchyma is generally absent in:
- (1) Dicotyledonous stems.
 - (2) Most of the monocotyledons
 - (3) Dicotyledonous roots.
 - (4) Gymnosperms.
134. Choose the **incorrect** statement.
- (1) A typical flower has four different kinds of whorls arranged successively on the swollen end of the stalk or pedicel, called pulvinus.
 - (2) The petiole help hold the leaf blade to light.
 - (3) The leaves of garlic are modified to store food.
 - (4) The flower is meant for sexual reproduction.
135. Choose the **correct** statement.
- (1) The epidermal tissue system is also called the fundamental tissue system.
 - (2) The outer walls of guard cells are thicker than those towards the stomatal pore.
 - (3) The conjoint vascular bundles usually have the phloem located only on the outer side of xylem.
 - (4) Companion cells are characteristic of gymnosperms.

136. Which of the following animals belongs to the class having a persistent notochord throughout their life?

- (1) *Balaenoptera*
- (2) *Delphinus*
- (3) *Scoliodon*
- (4) *Aptenodytes*

137. Which of the following animals exhibits true metameric segmentation?

- (1) Earthworm
- (2) *Ascaris*
- (3) *Hydra*
- (4) Mollusca

138. Which of the following statements is **incorrect** regarding the phylum Platyhelminthes?

- (1) They are acoelomate and exhibit organ-level organization for the first time in the animal kingdom.
- (2) Fertilization is internal, and development involves many larval stages in parasitic forms like *Fasciola*.
- (3) Flame cells are specialized cells that primarily help in digestion and food capture.
- (4) Some members, like *Planaria*, possess a high capacity for regeneration.

139. Which of the following statements about Aves are **correct**?

- A. All birds can fly as their forelimbs are modified into wings.
 - B. Long bones in birds are pneumatic and aid in making the skeleton light.
 - C. Birds possess additional digestive chambers like crop and gizzard.
 - D. Birds have dry skin with only a single oil gland at the tail base.
 - E. Respiration in birds occurs through lungs and is aided by air sacs.
- (1) B, C, D and E only
 - (2) A, B and C only
 - (3) A, C and E only
 - (4) B, D and E only

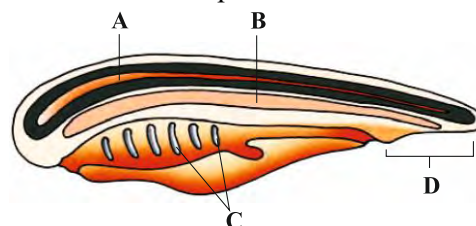
140. Match List-I with List-II.

List-I		List-II	
(A)	Annelids	(I)	Roundworms
(B)	Platyhelminthes	(II)	Polyp and medusa
(C)	Aschelminthes	(III)	Possess longitudinal and circular muscles
(D)	Coelenterata	(IV)	Flatworms

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-I, B-II, C-III, D-IV
- (3) A-III, B-IV, C-II, D-I
- (4) A-III, B-IV, C-I, D-II

141. Animals belonging to phylum chordata fundamentally characterised by the presence of structures noted as A, B, C, D, identify them and select the **correct** option



- (1) A- Notochord, B-Nerve cord, C - Gill slits, D- Post anal tail
- (2) A - Nerve cord, B - Notochord, C - Gill slits, D - Post anal tail
- (3) A - Nerve cord, B - Notochord, C Post anal - tail, D - Gill slits
- (4) A -Nerve cord, B Gill slits ,C -Notochord, D - Post anal tail

142. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Arthropods have been incredibly successful in colonizing almost every habitat on Earth.

Reason R: The presence of a versatile chitinous exoskeleton, jointed appendages for varied functions, and a well-developed tracheal tube system for direct oxygen supply to tissues are major contributing factors to their success.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R true and R is the correct explanation of A.
- (4) Both A and R true and R is not the correct explanation of A.

143. Read the following statements.

- A. They are exclusively marine, radially symmetrical, diploblastic organisms with tissue level of organisation.
- B. Body bears eight external rows of ciliated comb plates, which help in locomotion.
- C. Digestion is both extracellular and intracellular.
- D. Reproduction takes place only by sexual means.

Which of the following phylum is being described by above statements?

- (1) Platyhelminthes
- (2) Arthropoda
- (3) Mollusca
- (4) Ctenophora

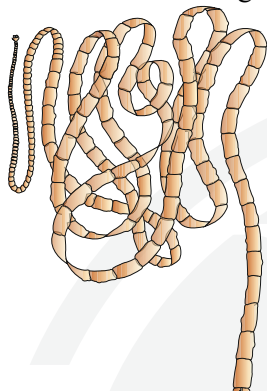
144. Read the following statements.

- A. In Urochordata, notochord extends from head to tail and is present throughout their life.
- B. Lancelet belongs to subphylum cephalochordata.
- C. Central nervous system is dorsal and hollow in Chordata.
- D. Chordata is divided into 3 subphyla: Hemichordata, Tunicata and Cephalochordata.

Which of the above statements are **true**?

- (1) B and C only (2) C and D only
- (3) A and C only (4) A and B only

145. Which of the following statements is **correct** regarding the animal shown in given figure?



- (1) It is diploblastic and acoelomate.
- (2) It shows true metamerism and segmentation.
- (3) It absorbs nutrients from the host directly through their body surface.
- (4) It has high regeneration capacity.

146. Consider the following characteristics:

- A. External fertilisation
- B. Dorso-ventrally flattened body
- C. Mostly endoparasites
- D. High regeneration potential in some
- E. Separate sexes

How many of the above are found in Platyhelminthes?

- (1) Only three (2) Only two
- (3) Only one (4) All five

147. During embryonic analysis, a scientist observed:

Species A: Could only be divided into mirror halves by a single plane, passing through the longitudinal axis.

Species B: Could be divided into similar halves through multiple planes passing through the central axis.

What is the **most** appropriate classification and an example for each?

- (1) Species A: Bilateral (e.g., Annelid), Species B: Radial (e.g., Ctenophore)
- (2) Species A: Radial (e.g., Coelenterate), Species B: Bilateral (e.g., Arthropod)
- (3) Species A: Asymmetrical (e.g., Porifera), Species B: Radial (e.g., Echinoderm)
- (4) Species A: Biradial (e.g., Ctenophore), Species B: Bilateral (e.g., Platyhelminth)

148. Which of the following sets of phyla contain animals having **true** coelom and bilateral symmetry in adult stages?

- (1) Platyhelminthes, Coelenterata and Ctenophora
- (2) Aschelminthes, Annelida and Arthropoda
- (3) Mollusca, Echinodermata and Platyhelminthes
- (4) Annelida, Arthropoda and Mollusca

149. Which of the following groups consists exclusively of cnidarians?

- (1) *Aplysia*, *Gorgonia*, *Spongilla*, *Hydra*
- (2) *Hydra*, *Pennatula*, *Obelia*, *Echinus*
- (3) *Hydra*, *Obelia*, *Adamsia*, *Physalia*
- (4) *Scypha*, *Hydra*, *Adamsia*, *Physalia*

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Roundworms are pseudocoelomates.

Reason R: In roundworms, the body cavity is not lined by mesoderm, instead, the mesoderm is present as scattered pouches in between the ectoderm and endoderm.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is not the correct explanation of A.
- (3) A is true but R is false.
- (4) A is false but R is true.

151. Which of the following statements about levels of organisation reflects a misconception?

- (1) The transition from the cellular level in sponges to the tissue level in cnidarians represents a major leap, allowing for coordinated functions like movement.
- (2) Organ level, as seen in Platyhelminthes, involves tissues organized into functional units like flame cells and eyespots.
- (3) In Aschelminthes, organs associate to form functional organ systems, such as a complete digestive system.
- (4) In Arthropods, the organ-system level is less advanced than in Molluscs because their cells are loosely arranged in aggregates.

152. Three embryos are analyzed:

Embryo X: Develops two germ layers and a non-cellular matrix between them.

Embryo Y: Develops three germ layers, but the body cavity is not lined by the third layer; rather, this layer exists as scattered pouches.

Embryo Z: Develops three germ layers, and a body cavity forms which is completely lined by the third layer.

What are the **correct** developmental classifications and coelomic conditions?

- (1) X: Diploblastic/Acoelomate; Y: Triploblastic/Pseudocoelomate; Z: Triploblastic/Euocoelomate
- (2) X: Triploblastic/Acoelomate; Y: Triploblastic/Euocoelomate; Z: Triploblastic/Pseudocoelomate
- (3) X: Diploblastic/Acoelomate; Y: Triploblastic/Acoelomate; Z: Triploblastic/Euocoelomate
- (4) X: Diploblastic/Pseudocoelomate; Y: Triploblastic/Pseudocoelomate; Z: Triploblastic/Euocoelomate

153. Match List-I with List-II.

List-I		List-II	
(A)	Fighting fish	(I)	<i>Pterophyllum</i>
(B)	Angel fish	(II)	<i>Betta</i>
(C)	Flying fish	(III)	<i>Pristis</i>
(D)	Saw fish	(IV)	<i>Exocoetus</i>

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-I, C-II, D-III
- (4) A-II, B-IV, C-I, D-III

154. Find the **incorrect** match.

- (1) Crow → *Corvus*
- (2) Pigeon → *Columba*
- (3) Parrot → *Psittacula*
- (4) Penguin → *Pavo*

155. **Statement I:** All Aschelminthes are exclusively parasitic in animals.

Statement II: Female roundworms are generally longer than the males.

Statement III: In Aschelminthes, body organisation is more advanced than that of Platyhelminthes.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both statement II & III are correct but statement I is incorrect
- (2) Only statement III is correct
- (3) Both statement I & statement III are correct
- (4) All statements are correct

156. Match List-I with List-II.

List-I (Level of organisation)		List-II (Animal)	
(A)	Organ level	(I)	<i>Pheretima</i>
(B)	Cellular aggregate level	(II)	<i>Fasciola</i>
(C)	Tissue level	(III)	<i>Spongilla</i>
(D)	Organ system level	(IV)	<i>Obelia</i>

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-II, B-IV, C-III, D-I
- (3) A-IV, B-I, C-II, D-III
- (4) A-I, B-IV, C-III, D-II

157. Which of the following sets of features is entirely **correct** for the human hookworm, *Ancylostoma duodenale*?

- (1) Acoelomate, monoecious, indirect development.
- (2) Pseudocoelomate, dioecious, males shorter than females, parasitic.
- (3) Euocoelomate, hermaphroditic, free-living.
- (4) Pseudocoelomate, dioecious, direct development, non-parasitic.

158. A biologist injected a dye into the circulatory systems of two animals: one Arthropod and one Annelid. In one animal, the dye spread in a confined path through vessels. In the other, it diffused into open cavities bathing the tissues directly. Which of the following is the **correct** conclusion?

- (1) Annelid has an open circulatory system; Arthropod has a closed one
- (2) Both animals possess closed circulatory systems
- (3) Arthropod has an open circulatory system; Annelid has a closed one
- (4) Both animals possess open circulatory systems

159. In an experimental setup, two sponges are placed in separate tanks:

Sponge A is in water containing fine food particles. **Sponge B** is in oxygen-rich but particle-free water. After some time, only Sponge A shows normal activity.

What conclusion can be drawn from this observation?

- (1) Sponges cannot utilize oxygen dissolved in water.
- (2) Sponges depend entirely on diffusion through their outer body wall.
- (3) The water current supplies both oxygen and food essential for their physiological activities.
- (4) Food particles are absorbed through the osculum, not through ostia.

- 160.** Examine the following statements about reptiles and identify the **correct** ones:
- Reptiles are homoiothermic vertebrates with four-chambered hearts.
 - Snakes and lizards cast off their skin along with scales.
 - All reptiles possess external ear openings represented by tympanum.
 - They are oviparous and development is direct.
 - Limbs, when present, are two pairs.
- B, C and D only
 - A, C and E only
 - A, B and D only
 - B, D and E only

- 161.** Match **List-I** with **List-II**.

List-I (Structure in sponges)		List-II (Function)	
(A)	Ostia	(I)	Single, large excurrent opening
(B)	Osculum	(II)	Central cavity lined by choanocytes
(C)	Spongocoel	(III)	Flagellated collar cells that generate water current and trap food particles
(D)	Choanocytes	(IV)	Tiny pores in the body through which water enters

Choose the **correct** answer from the options given below:

- A-I, B-IV, C-II, D-III
 - A-I, B-IV, C-III, D-II
 - A-IV, B-I, C-II, D-III
 - A-IV, B-II, C-I, D-III
- 162.** Select the **incorrect** option.
- Naja* - Cobra
 - Bangarus* - Krait
 - Vipera* - Viper
 - Hemidactylus* - Garden lizard
- 163.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: The heart is three chambered (one auricle and two ventricles) in frog.
Reason R: Circulatory system is closed type and frogs are cold blooded.
 In the light of the above statements, choose the **correct** answer from the options given below.
- A is true but R is false.
 - A is false but R is true.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true but R is not the correct explanation of A.

- 164.** Identify the aquatic mammal(s) from the following.
- Balaenoptera*
 - Equus*
 - Delphinus*
 - Pteropus*
 - Felis*
- I and III only
 - II and IV only
 - V only
 - IV and V only

- 165.** Given below are two statements:

Statement I: Mouth is located ventrally in the members of class Osteichthyes.

Statement II: The amphibian skin is without scales.
 In the light of the above statements, choose the most appropriate answer from the options given below.

- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.

- 166.** Study the given list of animals.

Pila, Pinctada, Sepia, Loligo, Octopus, Aplysia, Dentalium, Chaetopleura

How many of them are molluscs?

- Two
- Five
- Seven
- Eight

- 167.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: *Pristis* is a marine fish with cartilaginous exoskeleton.

Reason R: Mouth is located ventrally and notochord is persistent throughout life in *Pristis*.

In the light of the above statements, choose the **correct** answer from the options given below.

- A is true but R is false.
- A is false but R is true.
- Both A and R are true and R is the correct explanation of A.
- Both A and R are true but R is not the correct explanation of A.

- 168.** Complete the following analogy.

Pheretima : Nephridia :: *Saccoglossus* : _____

- Malpighian tubule
- Protonephridia
- Proboscis gland
- Green glands

169. Which of the following features is common to silverfish, scorpion, dragonfly, and prawn?

- (1) Three pairs of legs and unsegmented body.
- (2) Chitinous cuticle and two pairs of antennae.
- (3) Jointed appendages and chitinous exoskeleton.
- (4) Cephalothorax are present with tail.

170. Read the given statements and identify the animal and the phylum to which it belongs.

- A. The body consists of head, thorax and abdomen.
- B. It has jointed appendages.
- C. Tracheal tube system is present for respiration.
- D. Circulatory system is open type.
- E. It has economic importance.

- (1) *Pheretima* belongs to phylum Annelida.
- (2) *Apis* belongs to phylum Aschelminthes.
- (3) *Sepia* belongs to phylum Mollusca.
- (4) *Apis* belongs to phylum Arthropoda.

171. Given below are two statements:

Statement I: The most unique mammalian characteristic is the presence of milk producing glands or mammary glands.

Statement II: Air sacs connected to lungs supplement respiration in mammals.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

172. Match List-I with List-II.

List-I (Organism)		List-II (Significance)	
(A)	Locust	(I)	Vector for diseases like dengue and chikungunya
(B)	Butterfly	(II)	A gregarious pest causing crop damage
(C)	Prawn	(III)	Important pollinator for flowering plants
(D)	Mosquito	(IV)	Economically important as a food source

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-III, C-IV, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-III, C-I, D-IV

173. *Pavo* belongs to the class Aves whereas *Periplaneta* belongs to the largest phylum of kingdom Animalia.

The common feature among two of them is;

- (1) tendency to regulate body temperature.
- (2) presence of crop and gizzard.
- (3) presence of mesoderm in pouches forming pseudocoelom.
- (4) exoskeleton made up of polymer of N-acetyl glucosamine.

174. Select the **incorrect** match.

- (1) *Hydra* - fresh water polyp
- (2) *Branchiostoma* - protochordate
- (3) *Petromyzon* - jawless invertebrate
- (4) *Pristis* - 2-chambered heart

175. Match List-I with List-II.

List-I		List-II	
(A)	<i>Macropus</i>	(I)	Moist skin
(B)	<i>Struthio</i>	(II)	Hair on skin
(C)	<i>Hemidactylus</i>	(III)	Feathers on skin
(D)	<i>Rana</i>	(IV)	Dry cornified skin

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-III, D-I
- (2) A-IV, B-I, C-II, D-III
- (3) A-III, B-I, C-IV, D-II
- (4) A-II, B-III, C-IV, D-I

176. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Bioluminescence is well-marked in ctenophores.

Reason R: Ctenophores possess specialised cells that produce light due to biochemical reactions.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true and R is a correct explanation of A.
- (2) Both A and R are true but R is not a correct explanation of A.
- (3) A is true and R is false.
- (4) A is false and R is true.

177. Which of the following pairs are **correctly** matched?

- A. Poikilothermous – Birds and mammals
 - B. Agnatha – *Petromyzon* and *Myxine*
 - C. Tunicates – *Ascidia*, *Salpa* and *Doliolum*
 - D. Cartilaginous fishes – *Betta* and *Labeo*
- (1) A and B only
 - (2) A, B and C only
 - (3) B and C only
 - (4) A, B and D only

178. Choose the **incorrect** statement.

- (1) Statocysts are balancing organs present in aquatic arthropods.
- (2) A well developed digestive system is absent in echinoderms.
- (3) Sexual dimorphism is characteristic of organisms belongs to phylum Arthropoda.
- (4) The organisms of 2nd largest phylum exhibit same type of symmetry as that of larval forms of echinoderms.

179. Which of the following options is **correct** about reproduction in hemichordates?

- (1) Internal fertilisation.
- (2) Usually direct development.
- (3) Sexes are separate.
- (4) Monoecious

180. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Aschelminthes	(I)	This is the second largest animal phylum
(B)	Annelida	(II)	Often wormshaped, elongated
(C)	Arthropoda	(III)	Body segmentation like rings
(D)	Mollusca	(IV)	Exoskeleton of cuticle, jointed appendages

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-II, C-III, D-I
- (3) A-I, B-IV, C-III, D-II
- (4) A-II, B-III, C-IV, D-I





Test Analysis Sheet

- Student Name :
- PW Registration No. :
- Batch Code :
- Test Name/Type :

Max. Marks:
Marks Obtained:
Percentage:
Date of Exam :

1. Subject-wise Performance

Subject	Attempt Order	Time taken (min)	Marks Expected	No. of Attempted Q's	No. of Wrong Q's	Accuracy (%)
Physics						
Chemistry						
Botany						
Zoology						

- Was there a rush at the end? (Yes/No): _____
- Did you revise after solving? (Yes/No): _____

2. Wrong Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Calculation mistake					
Silly mistake					
Marked incorrect option, however solved correctly					
Conceptual Error					
Others					

3. Unattempted Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Forgot the concept / Formula					
Solved correctly but didn't have confidence to mark it on OMR					
Missed the class /Didn't revise					
Got confused with the concept					
Others					

4. Weaknesses Identified

(List topics/chapters where you lost marks)

- Physics :
- Chemistry :
- Botany :
- Zoology :

5. Action Plan Before Next Test

1. Types of Mistakes to Avoid:
2. Extra Practice Needed in:
3. Strategy Changes: (e.g., attempt Chemistry first, manage time better):
4. What would you like to improve emotionally/mentally for the next test? :

Student Signature: _____